Dissertation Learning Notes & Resources

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1 Reference Sheet

Force per unit area on fluid element: $\underline{F} = \int p\underline{n} \cdot d\underline{S} = \int_\Omega \nabla \cdot \mathbf{T}\, dV = \int_A \mathbf{T} \cdot \underline{n}\, dA$, where $\mathbf{T} \cdot \underline{n}$ is the stress on a surface normal to $\underline{n}$.

$T_{ij} = -p\delta_{ij} + 2\mu e_{ij}$ is the Stress Tensor. Components $\{i, j\}$ is the stress along i on the surface normal to j.

Material Derivative: $\frac{D}{Dt} := \frac{\partial}{\partial t} + (\underline{u} \cdot \nabla)$.

Navier Stokes Equations:

$$\nabla \cdot \underline{u} = 0, \tag{1a}$$

$$\underbrace{\rho\left(\frac{\partial \underline{u}}{\partial t} + \overbrace{(\underline{u}\cdot\nabla)\underline{u}}^{\text{inertial}}\right)}_{ma} = \underbrace{\overbrace{\mu\nabla^2\underline{u}}^{\text{viscous}} - \nabla p + \rho\underline{g}}_{\sum \underline{F}}. \tag{1b}$$

Stokes' Equations:

$$\nabla p = \mu\nabla^2\underline{u}, \tag{2a}$$

$$\nabla \cdot \underline{u} = 0. \tag{2b}$$

Streamlines: $\frac{\partial \underline{x}(s)}{\partial s} = \underline{u}(\underline{x}(s), t)$.

Reynolds Number: $\text{Re} = \frac{\rho U R}{\mu}$.

Bernoulli's Equations:

$$\tilde{B}(\underline{x}) = \left(\frac{\rho}{2}|u(x)|^2 + p + \rho g z\right), \qquad \text{(Steady, Inviscid. Constant on streamlines)} \tag{3a}$$

$$\hat{B}(t) = \left(\rho\frac{\partial \phi}{\partial t} + \frac{\rho}{2}|u(x,t)|^2 + p + \rho g z\right). \qquad \text{(Potential. Constant in space)} \tag{3b}$$

Potential / Irrotational Flows: $\underline{u} = \nabla\phi(\underline{x}, t) = \left(\frac{\partial\phi}{\partial r}, \frac{1}{r}\frac{\partial\phi}{\partial\theta}\right)$, with $\nabla^2\phi = 0$.

Incompressible Flows: $\underline{u} = \left(\frac{\partial\psi}{\partial y}, -\frac{\partial\psi}{\partial x}\right) = \left(\frac{1}{r}\frac{\partial\psi}{\partial\theta}, -\frac{\partial\psi}{\partial r}\right)$.

These link via: $\{\partial_x\phi = \partial_y\psi,\ \partial_y\phi = -\partial_x\psi\}$.

2 Problem Overview

Bounces are modelled specifically via the thin air film that coats the drop - lubrication theory. The drop bounces on it's *own air cushion*. Collisions with solids don't happen unless the solid is nano-smooth - small ridges puncture air film and reaches the liquid, sticking to it. Hence, drop-drop collisions (or symmetric to halve computation?).

The closer the drop gets to the barrier (and the thinner the air film gets), the more pressure forms. With no attractive forces, this pressure builds up until it repels the drop. At this initial phase, we are attempting to recreate figure 3a from Hicks and Pervis.

The drop is able to reach the barrier for two reasons: at small scales, the attract each other through van der Waals forces, and if the film height drops below the mean free path of gas molecules, pressure decreases, causing the gas molecules to slip along the boundary (Knudsen Gas-Kinetic Slip). If this happens, the the film ruptures and the drops merge.

Does this have something to do with the $1/r$ interaction distance from the summer reading?? Make notes on this!!

2.1 Key Assumptions

We don't really care about the droplet itself - it is high Reynolds number compared to the air film, much denser, and deforms not easily (surface tension acts perpendicular to the surface normal, and takes lots of energy to shape)

"Events before droplet splashing on a solid surface" says: The major simplifications for a typical millimetre-sized droplet with impact velocity of order $1ms^{-1}$ are (i) slender geometry for the gas film, (ii) the neglect of viscous forces in the liquid, (iii) the neglect of surface tension forces at the liquid–gas interface and (iv) the neglect of nonlinear inertia in the liquid. With respect to viscous forces, the liquid Reynolds number is of order $\mathrm{Re} = \rho RV/\mu \sim 1000$, where μ, ρ are the liquid viscosity and density respectively; hence inertial forces in the fluid initially overwhelm viscous forces, and we therefore neglect viscosity. Similarly, $\mathrm{We} = \rho V^2/\sigma \sim 100$ shows surface tension is negligible

. Combined with the fact that we initially model the drop's surface as a weak parabola (no 1st order terms), this means that we approximate the water's surface as locally smooth, and thus we can approximate the height of the film, $h(x)$ via linear functions. Similarly, at the simulation's middle (drop is closest to surface - pressure is largest - most of the maths and scaling happens here), we can assume that the drop's surface is locally flat (Plane-Poiseuille). Further assumptions on the liquid: inviscid and potential (Bernoulli, and $\nabla^2 p = 0$ - Laplace Equation, Green's function, etc).

On the other side, the gas is viscous-dominated rather than inertia dominated, and the height is much smaller than the x extent. Hence, we assume that the pressure does not change in y (Plane-Poiseuille), and that we can use the Lubrication equations. Assume the gas is a continuous fluid.

Hicks and Purvis further assume that we don't care about surface tension and gravity, and that we are incompressible. The former of these assumptions only really holds as the interface is smooth and flat, but this breaks down under high curvatures (interface wants to pull itself back together! HP doesn't go into this but we might!) The middle of these holds because something something Freud Number. The latter of these holds (CITE Mandre et al 2009) that this holds when $P_a \sqrt[3]{\frac{\mu_g}{\rho_l U R}} \gg 1$.

This seams really easy to surpass! - Thoroddsen et al (2005) and Dam and Le Clerc (2004) shows the impact velocity only needs to be more than $1ms^{-1}$! - This is okay though, as droplets in cloud formations have really slow collision velocities. If we really want to do this though, we modify the continuity equation in the lubrication equation, to consider a barotropic argument.

I think we're using the fact that, above a critical Reynolds Number (Smith et al. (2003) showed this is $\sim 10^7$), the interaction between air and liquid is purely inviscid. I think this means that the air cushion doesn't form? But it could also mean that the cushioning effect does not occur until the air layer is sufficiently thin? Ahhh, assume we are slow enough such that the pressure build-up isn't felt until late?

I am a bit confused about this in general... So the gas is considered to be viscous, but is this because of the thin film effect, or the interactions of a moving plate ramming into it? How does this change intuitively for show descent vs fast descent? It is very rare that we think about gas being anything other than INCREDIBLY inviscid!

HP shows that the Froude number is large, and therefore we neglect gravity.

2.2 Further Assumptions Worth Commenting on!

Heat transfer in the gas, mass diffusion from the gas into the liquid (we assume in our calculations that the gas layer cannot be absorbed into the liquid). Also, assume the gas is incompressible?? This seems a bit weird... see equation 2.3 in HP

From "Events before droplet splashing on a solid surface

2.3 Recap of Plane-Poiseuille Flow, and Lubrication Theory

The Plane-Poiseuille flow is the flow between two stationary plates of separation $h \sim \hat{\underline{z}}$.

- By the profile, we write $\underline{u} = u(y)\hat{\underline{x}}$. Via 1D nature, conservation of momentum gives no inertial terms, and:
  - x: $0 = -\frac{\partial p}{\partial x} + \mu \frac{\partial^2 u}{\partial y^2}$ *(friction moves from boundary into the fluid)*, y: $0 = -\frac{\partial p}{\partial y}$.
- As $p(x)$, $u(y)$, we just integrate to get $u(y) = \left(\frac{1}{2\mu}\frac{dp}{dx}\right) y^2 + Ay + B$.
- Using $u(y = \pm h/2) = 0$ (no-slip, viscous), we get the **parabola** $u(y) = \frac{1}{2\mu}\frac{dp}{dx}\left(y^2 - \frac{h^2}{4}\right)$.

Lubrication Theory: used as film radius $r \gg h$ height - very weak parabola. Hence, a fine assumption in this simple case is that the Poiseuille profile holds: $x, x^2 \to 0$ in the gas film, and we get a strip.

3 Techniques

The problem is basically trying to figure out the complex boundary conditions, given by the height of the drop's surface $h(x,t)$. We need the pressure at these points, as well as enforcing no-slip on the film. Matching these for both fluids gives a coupled 1D system.

In the drop, we are wanting to solve $\nabla^2 p = 0$, which means that the problem is determined purely by boundary conditions (ie: underbelly shape). We call these functions (p) **harmonic** - gradients blend smoothly towards the centre.

Methods of Note: Laplace equation, Half-Space Green's Functions, method of images, Dirichlet-to-Neumann (DtN) Maps, Cauchy PV, singularity subtraction, elliptical integrals

In the film, we are wanting so solve the lubrication equation, also known as Reynolds Equation. Via the smooth-ness of the drop's surface, and the fact that $h \ll R$, we can approximate h via linear functions at each point.

viscous shear against the wall dominates?? Hydraulic resistance scales as $1/h^3$, so it will eventually act like a stiff fluid?? What does this mean?

3.1 Green's Functions

A Dirac-Delta is a single 'prick' in space, which ripples outwards. Like how we can build a solution to

4 Hicks & Purvis Method

4.1 Intro

Weber number $\mathrm{We} = \frac{\rho R U^2}{\gamma}$, ratio of inertia (wants to deform) to surface tension (wants no deform) determines behaviour - middle ground only for bouncing.

We are modelling the shape of the drop's surface (from an initially parabolic shape). As the pressure increases as the drop descends, the drop is forced to flatten out over the surface (entropy - wants the pressure everywhere to be equal, but the surface is smooth). At the middle, where we have a large pressure spike, this 'pushing' forms a dimple! (imagine punching a balloon - the vast amount of the force is at the centre where the fist is, so this forces everything into a flat shape but pushes the inside more and deforms it! The difference here is that the pressure acts on the entire surface, but more so at the centre)

Get shape of pressure profile!! Does it follow a Gaussian?? Maybe this will help the analogy.

The droplet's fate is determined by this film: whether it bounces cleanly or ruptures the film.

This dimple is "rimmed by a razor-thin circular neck of minimum thickness h_{neck} (labelled h_{pd}). I have no idea what this means.

4.2 Non-Dimensionalisation

Using the scalings $p = \tilde{p} \times \rho_{\text{liquid}} U^2$ and $t = \tilde{t} \times R/U$, where R is the radius of the drop and U is the initial downward speed, we non-dimensionalise (1) and migrate to cylindrical coordinates to get:

Meseguer, A., & Trefethen, L. N. (2003). Linearized pipe flow to Reynolds number 107

$$\frac{1}{r}\frac{\partial (r u_r)}{\partial r} + \frac{1}{r}\frac{\partial u_\theta}{\partial \theta} + \frac{\partial u_z}{\partial z} = 0, \tag{4a}$$

$$\frac{\partial u_r}{\partial t} + (\underline{u}\cdot\nabla)u_r - \frac{u_\theta^2}{r} = -\frac{\partial p}{\partial r} + \frac{1}{\text{Re}}\left(\nabla^2 u_r - \frac{u_r}{r^2} - \frac{2}{r^2}\frac{\partial u_\theta}{\partial \theta}\right), \tag{4b}$$

$$\frac{\partial u_\theta}{\partial t} + (\underline{u}\cdot\nabla)u_\theta + \frac{u_r u_\theta}{r} = -\frac{1}{r}\frac{\partial p}{\partial \theta} + \frac{1}{\text{Re}}\left(\nabla^2 u_\theta - \frac{u_\theta}{r^2} + \frac{2}{r^2}\frac{\partial u_r}{\partial \theta}\right), \tag{4c}$$

$$\frac{\partial u_z}{\partial t} + (\underline{u}\cdot\nabla)u_z = -\frac{\partial p}{\partial z} + \frac{1}{\text{Re}}\nabla^2 u_z. \tag{4d}$$

4.3 Boundary Conditions

As for boundary conditions, impermeability and no-slip at the solid says

$$\underline{u}_g(r,0) = (0,0). \tag{5}$$

Applying this for the moving boundary at the interface gives the kinematic equation for impermeability: fluid particles on the boundary remain on the boundary . Defining the level set of the boundary $F(r,z,t) := z - h(r,t) = 0$, enforcing that the material derivative $\frac{DF}{Dt} = \left(\frac{\partial}{\partial t} + \underline{u}\cdot\nabla\right)F = 0$ evaluates directly to

CITE Lamb, H. (1932). Hydrodynamics and 1. Breslin JP, Andersen P. Hydrodynamics of Ship Propellers. Cambridge University Press; 1993

$$u_z(r,h) = \frac{\partial h}{\partial t} + u_r \frac{\partial h}{\partial r}. \tag{6}$$

This holds for both u_{gas} and u_{liquid}. and similarly no slip at the boundary. Also, from the lack of surface tension we get the normal stress condition $p_{\text{gas}} = p_{\text{liquid}}$ at the interface, using Bernoulli (potential flow, so we use (3b)).

This comes from matching $F = ma$ on the boundary (a dynamic boundary condition), and hence $\mathbf{T}_g \cdot \underline{\hat{n}} = \mathbf{T}_l \cdot \underline{\hat{n}}$. As the liquid is inviscid, $\mathbf{T}_l = -p\mathbf{I}$, and so we have $p = p$.

To add the effect of surface tension, we just say $p_g = p_l + ...$!

cite "Long-

Crucially, as the liquid is in the high Reynolds number regime (ie: the velocity $\underline{u} = \nabla\phi$, for some potential ϕ), we cannot enforce no slip for the liquid at the interface... BUT WHAT ABOUT THE GAS!?!? We have the kinematic equation, but doesn't this mean that the tangential velocity of the gas moves the same as the tangential velocity of the gas? And therefore, no slip? But the boundary is controlled by the gas, so no?? Maybe we do get slip? Talk about boundary layer effect in the report? (as it is a liquid-liquid interface vs a liquid-solid interface? Is it just the difference in viscosity, or the absolute magnitude of the liquid viscosity)

Okay, here's what's going on: the liquid is inviscid, so we lose slip. Hence, as we are dealing with a fluid-fluid interface, NEITHER fluid can have no-slip conditions. We only enforce impermeability via the lack of boundary conditions available from Laplace's equation. However, the moving interface creates the kinematic boundary equation (forces a streamline) and pressure matching boundary condition. This is made simpler as $h(r,t)$ is not a one-to-many function?

And the boundary conditions $\underline{u}(r,0) = (0,0)$, $u_z(r,h) = \frac{dh}{dt} + u_r\frac{dh}{dr}$, $\underline{u}(r,z) = \underline{u}_{\text{liquid}}$, $p = p_{\text{liquid}}$ at the interface.

4.4 Lubrication Equation Derivation

By the axisymmetric conditions, $\partial/\partial\theta = 0$ and $u_\theta = 0$, and thus we get $u(r,z)$. Hence, in the absence of external forces (including gravity), we get:

From "Stability of nonswirl axisymmetric solutions to the Navier-Stokes equations"

$$\frac{1}{r}\frac{\partial(ru_r)}{\partial r} + \frac{\partial u_z}{\partial z} = 0, \tag{7a}$$

$$\rho_g\left(\frac{\partial u_r}{\partial t} + u_r\frac{\partial u_r}{\partial r} + u_z\frac{\partial u_r}{\partial z}\right) = -\frac{\partial p_g}{\partial r} + \mu_g\left[\frac{\partial}{\partial r}\left(\frac{1}{r}\frac{\partial(ru_r)}{\partial r}\right) + \frac{\partial^2 u_r}{\partial z^2}\right], \tag{7b}$$

$$\rho_g\left(\frac{\partial u_z}{\partial t} + u_r\frac{\partial u_z}{\partial r} + u_z\frac{\partial u_z}{\partial z}\right) = -\frac{\partial p_g}{\partial z} + \mu_g\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u_z}{\partial r}\right) + \frac{\partial^2 u_z}{\partial z^2}\right]. \tag{7c}$$

Rescaling the momentum equations under the regime of $h/R \ll 1$ (ie: locally flat), we conclude that, to leading order:

cite Schlichting, 1968, Long-scale evolution of thin liquid films

$$\frac{\partial p_g}{\partial r} = \mu_g\frac{\partial^2 u_r}{\partial z^2}, \tag{8a}$$

$$\frac{\partial p_g}{\partial z} = 0. \tag{8b}$$

This directly concludes that p_g is a function of r and t only! Using the boundary conditions of $\underline{u}(r,0) = \underline{u}(r,h(r,t)) = 0$, we can integrate (8a) to get the Plane Poiseuille velocity profile :

CITE: Long-scale evolution of thin liquid films

$$u_r(r,z,t) = \frac{z(z-h(r,t))}{2\mu_g}\frac{\partial p_g(r,t)}{\partial r}, \tag{9}$$

with velocity flux,

$$Q(r,t) = \int_0^{h(r,t)} u_r\,dz = -\frac{h^3}{12\mu_g}\frac{\partial p_g(r,t)}{\partial r}. \tag{10}$$

To determine the evolution of the film thickness $h(r,t)$, we integrate the continuity equation (7a) inside the entire z domain to get,

CITE NOTES 2 DERIVATION OF THE CLASSICAL REYNOLDS EQUATION FOR THIN

$$\int_0^{h(r,t)} \frac{1}{r}\frac{\partial(ru_r)}{\partial r}\,dz + \int_0^{h(r,t)} \frac{\partial u_z}{\partial z}\,dz = 0. \tag{11}$$

Recalling Leibniz Integration Rule, and applying (6), the first term in (11) evaluates to

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\int_0^h u_r\,dz\right) - u_r(r,h)\frac{\partial h}{\partial r} = \frac{1}{r}\frac{\partial(rQ)}{\partial r} - \left(u_z(r,h) - \frac{\partial h}{\partial t}\right), \tag{12}$$

whilst the Fundamental Theorem of Calculus and boundary condition (5) can be used to evaluate the second term in (11) as merely $u_z(r,h)$. Combining this with (12) in continuity equation (11), whilst subbing in (10) for Q, we retrieve the axisymmetric Reynolds lubrication equation :

$$\frac{\partial h(r,t)}{\partial t} = \frac{1}{r}\frac{\partial}{\partial r}\left(\frac{rh^3}{12\mu_g}\frac{\partial p_g(r,t)}{\partial r}\right). \tag{13}$$

Scaling radial lengths by the drop radius R, film thickness by H, time by R/U, and pressures by the liquid inertial scale $\rho_l U^2$, this can be expressed in non-dimensional form as:

$$\frac{\partial h(r,t)}{\partial t} = \frac{1}{r}\frac{\partial}{\partial r}\left(\frac{\mathrm{St}}{12}rh^3\frac{\partial p_g(r,t)}{\partial r}\right), \tag{14}$$

where $\mathrm{St} = \frac{\rho_l RU}{\mu_g}$ is the Stokes number.

4.5 Droplet Equations Formation

We introduce the characteristic pressure length ℓ (the horizontal length that pressure makes a leading order contribution to the deformation of the drop), or rather the ratio $\delta := \ell/R$. Rescaling with this ($L \to \delta L$, $t \to \delta^2 t$, $h \to \delta^2 h$, $U \to U$ (and thus other components of drop velocity! Don't really care about oblique impacts unless they are VERY oblique - solutions match to leading order), $p \to p/\delta$), we approximate the interface as flat to leading order. We recover that as $z \to 0$, the material derivative $\frac{D}{Dt}$ loses it's advective terms , and thus by the high Reynolds Number approximation, the axisymmetric version of (4) becomes

$$\frac{\partial \underline{u_l}}{\partial t} = -\left(\frac{\partial p_l}{\partial r}, \frac{\partial p_l}{\partial z}\right), \tag{15}$$

and so taking the cylindrical divergence of (15), and equating to (7a) gives ,

$$\nabla^2 p_l = 0. \tag{16}$$

In other words, the pressure field is Harmonic within the liquid. By the weak maximum principle for the Laplace Equation, we can therefore determine p uniquely from the boundary conditions (for example, using the Dirichlet Principle)

Possible extension for visuals and post processing: create big 3D heatmap of the pressure inside the liquid drop (we don't really care about the details and pressure inside the drop, but it's useful as a learning tool since we have the data). Requires spacial drop discretisation.

To reduce this to a boundary integral, which can be solved to generate an ODE for $h(r,t)$, we use the concept of Green's functions. Intuitively, these are the responses to Dirac Delta functions Green's Second Identity states

$$\iiint_\Omega \left(\phi\nabla'^2 G - G\nabla'^2\phi\right) r'\,dr'\,d\theta'\,dz' = \iint_{\partial\Omega}\left(\phi\frac{\partial G}{\partial n'} - G\frac{\partial \phi}{\partial n'}\right) dS' \tag{17}$$

4.6 Formation of the Problem

Now we have non-dimensionalisd, we turn to the symmetries that allow for a simpler argument. The whole problem is axisymmetric in the z axis, hence we can perform the maths on the 2D version of the problem, then convert to a 3D problem by integrating the solution over θ. Hence, it

CITE: Modelling and closed loop control of near-field acoustically levitated objects, A rotational travelling wave based levitation device - Modelling, design, and control

CITE: HP, Air cushioning with a lubrication/inviscid balance

CITE: HP, Air cushioning with a lubrication/inviscid balance

cite: Evans, L. C. (2010). Partial Differential Equations

CITE: Partial Differential Equations: An Intro-

is natural to use polar coordinates (r, z) for this: if we used (x, y), the geometry would represent a infinitely long cylinder, rather than a circular drop.

Papers on 2D problem, for comparison: Mandre, Mani & Brenner, Precursors to splashing of liquid droplets on a solid surface

Again, we can consider normal impact as even oblique impacts are normal to leading order! Also, assume that the initial pressure in the gas is $p = 0$.

4.7 Numerics Time Yay!

The lubrication equation (14) and boundary integral formula (??) couple to form an iterative method. Given the initial condition of the downward moving drop in the far field, we update the pressure via (14), then use this pressure to update the interface via (??), etc etc.

Something about picking a predictor step?

5 Analysis & Numerics

For an analytic comparison, we can compare the $\mathrm{St}^{-1/3}$ dimple scaling.

5.1 Asymptotics

5.2 Figures

Use a Kymo Graph! :D

6 Additions to Purvis-Hicks

Disjoining Pressure (microscopic van der Waals attraction) pulls interfaces together. Knudsen Gas-Kinetic Slip (as h goes to mean free path, the cushion strength is reduced as the molecules begin to slip). The scaling is achieved by the Kn number. As the drop descends, the following things happen in order:

1. We get slip at the boundaries.

   Boundary Layer Theory: on the liquid side of the interface, we get a boundary layer like in MD3D1, but on the gas side (which we model as viscous), it is kinetic effects that give boundary layers! Ie: as $h \to 0$, we lose this kinetic effect and thus slip!

2. The bulk equations begin to break.

To resolve this, we consider viscosity $\mu \to \mu(h)$ at these small scales.

7 Report Layout

7.1 Introduction

Say how we can discuss the 3D problem using the language of axi-symmetric-ness and therefore turn things into a 2D problem.

7.2 Work to Delegate to Jiya

Non dimensionalisation and scaling for $St^{-1/3}$, Asymptotic scale separation, ...

8 Handwritten Uploads

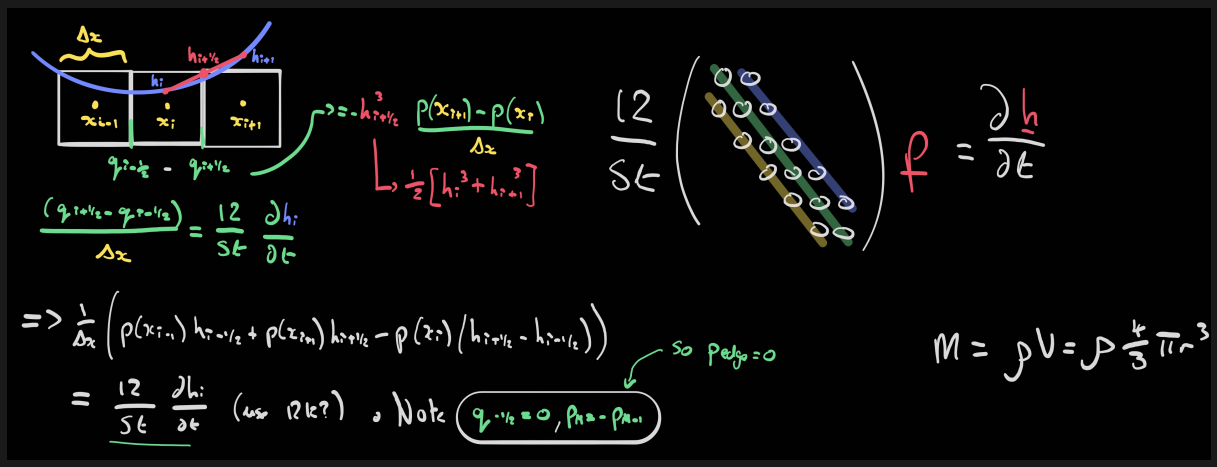


Figure 1: 13.10.2026 : Full domain discretisation and matrix construction.

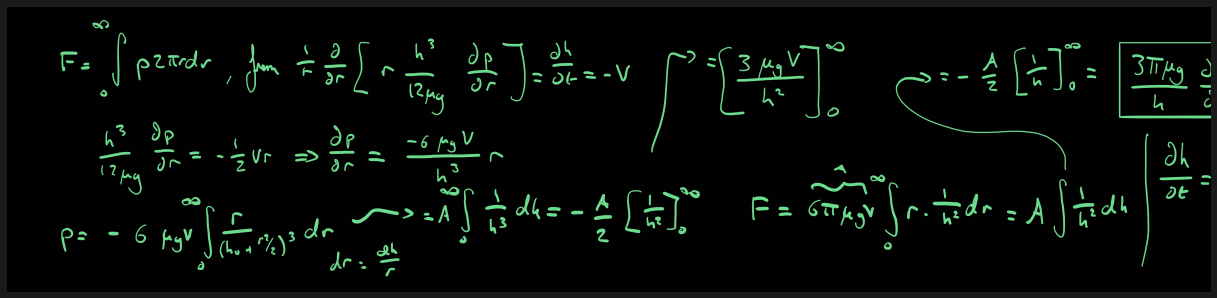


Figure 2: 13.10.2026 : Derivation of forces acting on the interface (from the air cushion pressure).

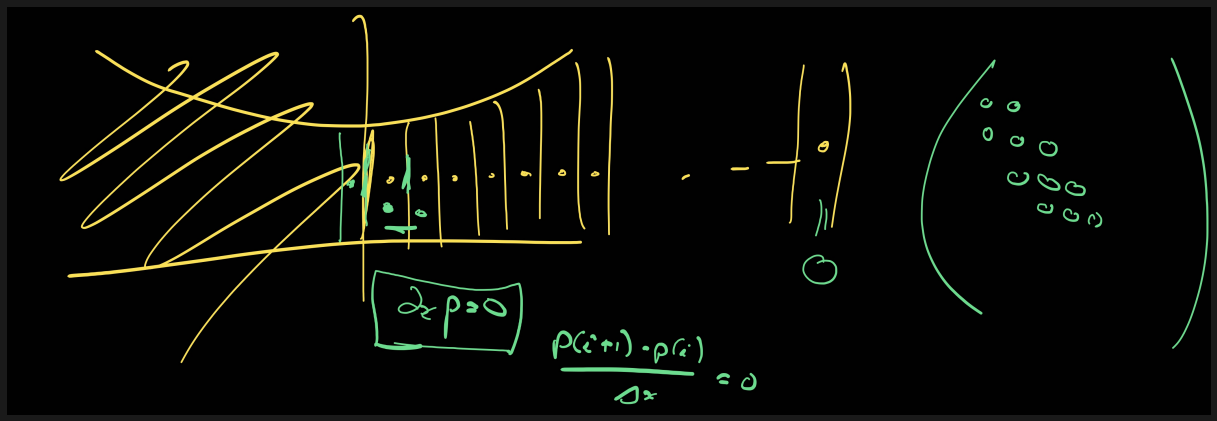


Figure 3: 13.10.2026 : First example discretising the domain.

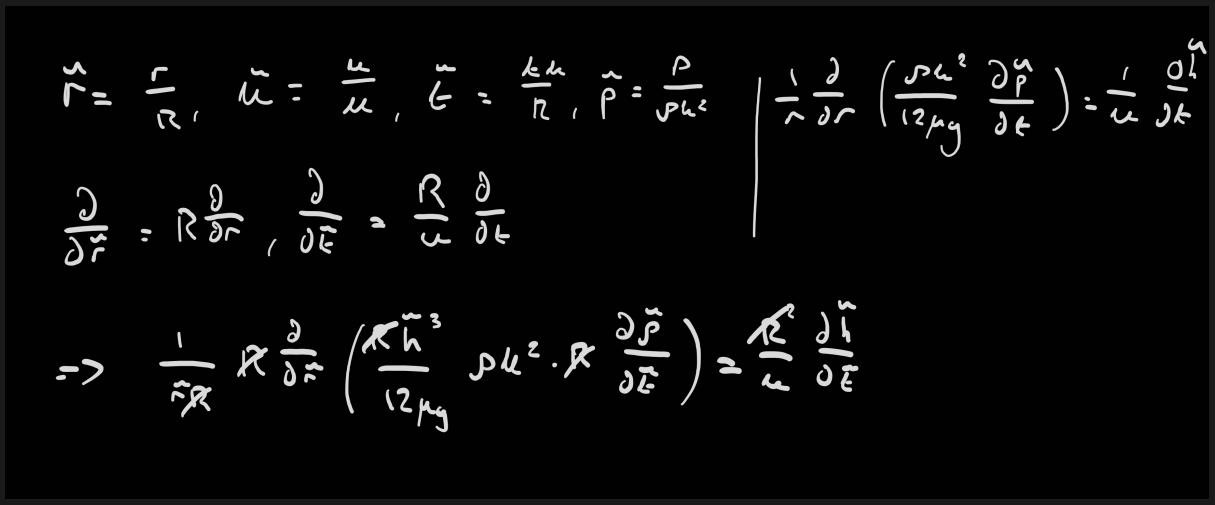


Figure 4: 13.10.2026 : Non-dimensionalisation of the Lubrication Equation.